



basic education

Department:
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SENIOR CERTIFICATE EXAMINATIONS/ NATIONAL SENIOR CERTIFICATE EXAMINATIONS

INFORMATION TECHNOLOGY P2

MAY/JUNE 2026

MARKS: 150

TIME: 3 hours

This question paper consists of 17 pages.

INSTRUCTIONS AND INFORMATION

1. This question paper consists of SIX sections:

SECTION A:	Short Questions	(20)
SECTION B:	Systems Technologies	(26)
SECTION C:	Communication and Network Technologies	(25)
SECTION D:	Data and Information Management	(25)
SECTION E:	Solution Development	(26)
SECTION F:	Integrated Scenario	(28)
2. Read ALL the questions carefully.
3. Answer ALL the questions.
4. The mark allocation generally gives an indication of the number of facts/reasons required.
5. Number the answers correctly according to the numbering system used in this question paper.
6. Write neatly and legibly.

SECTION A: SHORT QUESTIONS**QUESTION 1**

- 1.1 Various options are provided as possible answers to the following questions. Choose the answer and write only the letter (A–D) next to the question numbers (1.1.1 to 1.1.10) in the ANSWER BOOK, e.g. 1.1.11 D.

- 1.1.1 Which ONE of the following is NOT a transmission medium?
- A NFC
 - B UTP
 - C Fibre optic
 - D Infrared
- (1)
- 1.1.2 The main goal of SEO is to ...
- A increase the number of results of a search.
 - B create and design a website for faster loading times.
 - C improve a website's visibility on a search engine.
 - D protect a website from cyber threats.
- (1)
- 1.1.3 ... is a popular communication standard that is primarily used for video and audio output to television sets and monitors.
- A VGA
 - B PCI
 - C USB 2.0
 - D HDMI
- (1)
- 1.1.4 A language that is used to enhance the formatting and layout of a website:
- A AJAX
 - B PHP
 - C CSS
 - D JavaScript
- (1)
- 1.1.5 Which ONE of the following is NOT an example of a web browser?
- A Google
 - B Microsoft Edge
 - C Chrome
 - D Safari
- (1)

1.1.6 Consider the following e-mail that was sent:

To	jane@hypermail.com
CC	michael@acme.com
BCC	siya@securemail.com; peter@securemail.com
Subject	Party next weekend!
Body	Hi everyone! Party at my place next week at 20:00. See you there! Regards, Rob

Which ONE of the following statements is TRUE?

- A Jane will know that Michael and Siya were invited.
- B Siya will know that Michael and Jane were invited.
- C Michael will know that Siya was invited.
- D Siya will know that Peter was invited. (1)

1.1.7 Which ONE of the following is a networking device that is used to connect multiple wireless devices to a single wired network?

- A Access Point
- B ADSL
- C Switch
- D Hub (1)

1.1.8 The unit of measurement used for the speed of data transferred to and from an HDD is ...

- A gigahertz (GHz).
- B megabits per second (Mbps).
- C gigabits per second (Gbps).
- D megabytes per second (MBps). (1)

1.1.9 Which keyword is used in an SQL statement to filter records of a database table using a condition?

- A FORMAT
- B UPDATE
- C WHERE
- D LENGTH (1)

1.1.10 Determine the integer value of the variable **iCalculation** when the following Delphi statement is executed:

`iCalculation := 15 MOD 3 + 5 DIV 2 - Ceil(11/4)`

- A 3
- B -1
- C 5
- D 0 (1)

1.2 Give ONE word/term for each of the following descriptions. Write only the word/term next to the question numbers (1.2.1 to 1.2.5) in the ANSWER BOOK, e.g. 1.2.6 Computer.

- 1.2.1 Very large data sets that are analysed using computers to reveal trends and associations (1)
- 1.2.2 An electronic document that serves to authenticate the security and identity of a website (1)
- 1.2.3 A legal contract between the owner and the user of the software (1)
- 1.2.4 The set of characters that appear after the dot in a filename and indicates a file's format or the type of file (1)
- 1.2.5 A processing technique where the CPU switches rapidly between tasks, creating the illusion of running the tasks simultaneously (1)

1.3 Indicate whether the following statements are TRUE or FALSE. Choose the answer and write 'true' or 'false' next to the question numbers (1.3.1 to 1.3.5) in the ANSWER BOOK. Correct the statement if it is FALSE by changing the underlined word/phrase.

NOTE:

- Do not simply use the word 'NOT' to change the statement.
- NO marks will be awarded if only FALSE is provided as the answer.

- 1.3.1 Copyleft is a license agreement under which a user may copy, modify and distribute an author's work, as long as the same rights are preserved in the modified work. (1)
- 1.3.2 An OTP (one-time pin) is sent to a user's smartphone via SMS or e-mail for the purpose of authenticating the user. (1)
- 1.3.3 Broadband is the maximum amount of data that can be transmitted, in a specific time period, from one point to another in a network. (1)
- 1.3.4 LAN refers to the physical arrangement of the hardware in a network. (1)
- 1.3.5 The hexadecimal value E₁₆ is equal to the decimal value 13. (1)

TOTAL SECTION A: 20

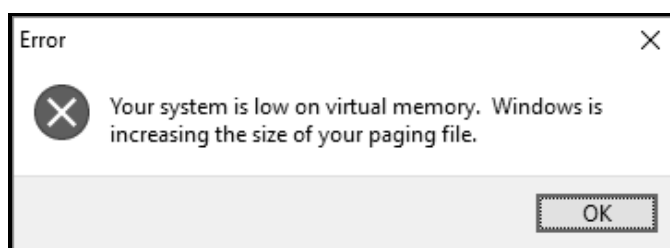
SECTION B: SYSTEMS TECHNOLOGIES**QUESTION 2****SCENARIO**

CompuMart, a leading computer shop, is implementing improvement strategies to provide better services to both in-store and online customers.

2.1 The CompuMart team evaluates the latest memory and storage options of the computers at the store.

2.1.1 Differentiate between a *USB flash drive* and a *hard disk drive (HDD)* in terms of the technology used for data storage. (2)

2.1.2 While working on one of the computers in the testing lab, the following error message appeared:



(a) Briefly explain why an operating system would report a low virtual memory problem in terms of memory management. (2)

(b) Name ONE immediate consequence of low virtual memory if the warning is ignored and the problem is NOT fixed by increasing the paging file. (1)

2.2 The technicians at CompuMart are required to advise their customers on upgrading monitors.

2.2.1 Customers often confuse the size of a monitor with the resolution and the refresh rate of the monitor.

Distinguish between the THREE monitor-related concepts (size, resolution and refresh rate) by briefly explaining what EACH term refers to:

(a) Size (1)

(b) Resolution (1)

(c) Refresh rate (1)

- 2.2.2 Gaming is a popular activity among the CompuMart customers.
- (a) Motivate why a monitor with a high refresh rate and low response time would be recommended for gaming purposes.
- Specifically state the advantage of EACH of the TWO features (high refresh rate and low response time) for gaming purposes as part of your answer. (2)
- (b) The graphics processing unit (GPU) of a computer system is either integrated on the CPU (iGPU) or it can reside on a video card which slots into a PCIe slot on the motherboard.
- Motivate why a GPU on a video card is the better option for graphic intensive software usage, such as gaming. (2)
- 2.3 CompuMart staff offer customers the option to update the BIOS settings on their computers and to upgrade or replace the CMOS battery.
- 2.3.1 Explain the relationship between BIOS and CMOS. (2)
- 2.3.2 Give TWO reasons why customers are advised to update the BIOS settings of a computer system. (2)
- 2.3.3 State ONE potential risk of updating the BIOS of a computer system. (1)
- 2.4 CompuMart has introduced cloud services to offer customers secure remote storage, backup solutions and other cloud services, such as Software as a Service (SaaS).
- 2.4.1 State TWO advantages that SaaS provides over traditional software licencing models. (2)
- 2.4.2 State TWO disadvantages of using cloud services for storage purposes. (2)
- 2.5 The service desk at the company provides advice and information to customers on hardware, software and power management solutions.
- 2.5.1 Briefly explain what the purpose of a device driver is. (1)
- 2.5.2 Give TWO examples of power management settings, in an operating system, that can contribute to energy savings. (2)

2.5.3 Write down TWO types of system software from the list below.

- Word processor
- Device drivers
- Backup software
- Spreadsheets
- Operating system
- Presentation software

(2)

TOTAL SECTION B: 26

SECTION C: COMMUNICATION AND NETWORK TECHNOLOGIES**QUESTION 3****SCENARIO**

CompuMart employs professional consultants who are highly qualified in network technologies to better serve their customers.

- 3.1 Answer the following frequently asked questions to assist customers in understanding networking concepts:
- 3.1.1 Why would a computer network better serve CompuMart, rather than traditional stand-alone PCs? (2)
 - 3.1.2 Briefly explain why a network operating system (NOS) is NOT required for a peer-to-peer network. (1)
 - 3.1.3 Which communication component is required for a computer to connect to a home area network (HAN)? (1)
 - 3.1.4 Which protocol allows for telephone calls to be made over the internet? (1)
 - 3.1.5 Name the protocol most commonly used to upload/download files to/from the internet. (1)
- 3.2 The store provides a public Wi-Fi hotspot as part of the service that they provide to their customers.
- 3.2.1 Explain what a *Wi-Fi hotspot* is. (2)
 - 3.2.2 State TWO risks associated with the use of a public Wi-Fi hotspot. (2)
- 3.3 The store decided to introduce remote desktop support to assist customers with software issues.
- 3.3.1 Give TWO reasons to motivate why the store introduced remote desktop support rather than on-site support. (2)
 - 3.3.2 The technician providing remote desktop support will need the IP address of the customer's desktop to establish a connection.
 - (a) Explain what an *IP address* is. (2)
 - (b) State ONE possible disadvantage of using a remote desktop connection. (1)

- 3.4 Different network communication standards can be used to set up networks, which affect the quality and speed of communication.

Arrange the following network standards in the correct order from the slowest to the fastest based on their typical speed:

3G; 5G; Edge; LTE (2)

- 3.5 A few customers who required software to be installed on their computers have chosen not to purchase proprietary software. They claim that the proprietary software that they require is downloadable free of charge from BitTorrent.

Critically discuss the legality of the customers' claim. (2)

- 3.6 CompuMart emphasises data security as a critical matter for customers to attend to.

3.6.1 Explain the role of a public key and a private key in SSL encryption. (2)

3.6.2 Discuss TWO functions of a firewall in network security. (2)

3.6.3 Motivate why a VPN is considered secure for browsing purposes. (2)

TOTAL SECTION C: 25

SECTION D: DATA AND INFORMATION MANAGEMENT**QUESTION 4**

- 4.1 The CompuMart team is conducting research on the most suitable database management system (DBMS) for the company, which has branches all over the country.

4.1.1 What is the purpose of a DBMS? (2)

4.1.2 Motivate the implementation of a database management system (DBMS) by stating ONE benefit of having a DBMS. (1)

4.1.3 The table below shows a comparison between a distributed database system and the server (centralised) database system in terms of data storage and system reliability.

Copy statements (a) to (d) in the table below into the ANSWER BOOK and complete each statement correctly.

FEATURE	DISTRIBUTED DATABASE SYSTEM	SERVER DATABASE SYSTEM
Storage of data	(a) Data is stored ...	(b) Data is stored ...
System reliability	(c) More reliable because ...	(d) Less reliable because ...

(4)

- 4.2 It is important for a business to apply transaction tracking in a database.

The information in the table below is an example of transaction log entries of actions that took place in a database.

TransID	UserID	StartTime	EndTime	Action	Status
T101	U12	10:15:02	10:15:07	Add new customer	Completed
T102	U15	10:15:10	10:15:14	Update stock	Rollback
T103	U12	10:15:15	10:15:16	Process payment	Completed
T104	U20	10:15:20	10:15:29	Delete order	Rollback

4.2.1 What would be the result of rolling back transaction T102? (1)

4.2.2 Give ONE possible reason why a rollback needs to be done on a transaction in a database. (1)

4.2.3 Use the information in the table above to explain TWO benefits of keeping transaction log entries available as part of database management. (2)

- 4.3 Validation and verification are two techniques that are used to ensure that data being captured is reliable.

Briefly explain the difference between *validation* and *verification* by explaining what EACH of these techniques does.

(2)

- 4.4 The CompuMart company is considering the possibility of expanding their operations. A decision support system (DSS) could be useful in their venture to grow and expand their business.

Briefly explain what a *decision support system* is.

(1)

- 4.5 A relational database can be used to store a large amount of interrelated data.

4.5.1 State TWO advantages of using a relational database.

(2)

4.5.2 The database used in the CompuMart store includes the following tables and fields:

tblCustomers	tblPurchases	tblProducts
CustomerID	PurchaseID	ProductID
CustomerName	PurchaseDate	ProductName
E-mail	CustomerID	Description
CellNumber	ProductID	Price
Address	Quantity	StockQuantity

- (a) Define the term *field*. (1)
- (b) What is the role of the foreign key field **CustomerID** in the **tblPurchases** table? (2)
- (c) Name TWO possible alternate keys in the **tblCustomers** table. (2)
- (d) Design a simple ERD (entity-relationship diagram) illustrating the structure and relationship between the **tblProducts** and **tblPurchases** tables. (4)

TOTAL SECTION D: 25

SECTION E: SOLUTION DEVELOPMENT**QUESTION 5**

- 5.1 Evaluate the statements in the segment of code below and answer the questions that follow.

Statements
for iCnt := 1 to 15 do;
if iCnt MOD 5 = 0 then
redOutput.Lines.Add(IntToStr(iCnt));
ShowMessage(IntToStr(iCnt));

- 5.1.1 Explain why the expected output, which is to display the multiples of 5 from 1 to 15, will NOT be displayed. (2)
- 5.1.2 What value will be displayed in the ShowMessage dialogue box? (1)
- 5.2 Searching for an element in an array can be done using the binary search technique. The declaration of two arrays, **arrNum1** and **arrNum2**, have been provided below.
- arrNum1 : array[1..7] of Integer = (12,15,17,34,78,83,92);
arrNum2 : array[1..7] of Integer = (92,17,78,15,12,83,34);
- 5.2.1 Which ONE of the arrays provided above will be more suitable to use to implement the binary search technique? (1)
- 5.2.2 Give a reason for your answer to QUESTION 5.2.1. (1)
- 5.2.3 One of the steps in the algorithm of the binary search technique is to locate the element at the middle index/position in an array.
- Write a Delphi statement that will determine the middle index of an array called **arrNumbers** with an unknown number of elements. Assign the middle index value to a variable called **iMid**. (2)

- 5.3 The Delphi code provided below must be evaluated and traced using a trace table.

Line No.	Delphi code
	var
	sOut : String;
	iCnt, iLoop : Integer;
	begin
1	sOut := '#';
2	for iLoop := 1 to 3 do
	begin
3	iCnt := 1;
4	while iLoop <> iCnt do
	begin
5	sOut := sOut + '*';
6	Inc(iCnt);
	end;
7	redOutput.Lines.Add(sOut);
	end;
	end;

Redraw the trace table structure below into the ANSWER BOOK and complete the trace table using the Delphi code provided above.

NOTE: You are allowed to add more rows to the trace table.

Line No.	sOut	iLoop	iCnt	iLoop <> iCnt ?	Output

(6)

- 5.4 A text file called '**Names.txt**', which contains names, is provided in an existing Delphi project folder.

Study the segment of Delphi code provided below.

```
Assignfile(tFile, 'Names.txt');
```

```
Try
```

```
  (x);
```

```
Except
```

```
  (y);
```

```
End;
```

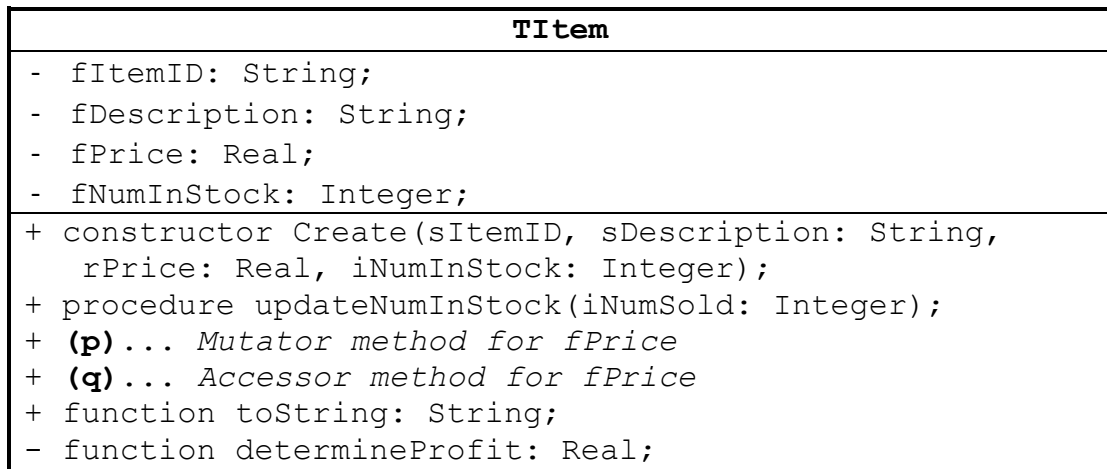
- 5.4.1 Write down the Delphi statement that must replace **(x)** in the code above to prepare the existing text file **Names.txt** to add a name to the file.

(1)

- 5.4.2 When will the statement(s) represented by **(y)** be executed?

(1)

- 5.5 The following UML class diagram which describes an **Item** object has been provided:



- 5.5.1 Complete the UML class diagram with statements as described below.

- (a) Write a statement to replace **(p)** with the heading of a mutator method for the **fPrice** attribute. The method receives a value to be assigned to **fPrice** as a parameter. (3)
- (b) Write a statement to replace **(q)** with the heading of an accessor method for the **fPrice** attribute. (3)

- 5.5.2 A global object, **objItem**, has been declared in a form class that is linked to the **TItem** object class.

- (a) The following statement is an attempt to instantiate the object **objItem** using the constructor create method as described in the given UML class diagram:

```
TItem := objItem.create(sDescription, rPrice, iNumInStock);
```

State TWO errors in the statement above. (2)

- (b) Assume that the object **objItem** has been instantiated correctly. Why will the following statement in the form class return an error?

```
rProfit := objItem.determineProfit;
```

(1)

- 5.5.3 Assume that the provided parameterised constructor in the UML diagram was replaced by the following default constructor method:

```
Constructor Create();
```

State the changes that will need to be effected to the UML class diagram to ensure that all attributes of the **TItem** object can still be assigned with values. (2)

TOTAL SECTION E: 26

SECTION F: INTEGRATED SCENARIO**QUESTION 6****SCENARIO**

The CompuMart store is investing in state-of-the-art hardware components, updating its system software and enhancing network connectivity. The management is focused on ensuring that these upgrades are secure, efficient and scalable to meet the evolving demands of a competitive retail market.

- 6.1 At CompuMart's repair and service desk, regular maintenance is carried out to keep systems running smoothly.
- 6.1.1 Name the utility software that is available as part of Microsoft Windows, that is used for removing temporary internet files and downloads. (1)
- 6.1.2 State TWO potential benefits of a *custom software installation* versus a *full software installation*. (2)
- 6.1.3 State ONE benefit of performing regular system backups. (1)
- 6.2 CompuMart's management is investigating various techniques to increase the sales at the store.
- 6.2.1 The company wants to improve the experience of customers by reducing the time that customers spend at the store while waiting to be served by staff.
- Name AND motivate the implementation of TWO devices that can be used to allow customers to purchase items without staff assistance. (4)
- 6.2.2 Describe TWO ways in which social media websites may be used to influence sales at the store. (2)
- 6.3 Customers often call the store to inform them that their computer system has been hacked or infected by malware.
- 6.3.1 A rootkit is one of the common types of malware that customers should be aware of.
- Explain what a *rootkit* is. (2)
- 6.3.2 CompuMart is running a program known as CAPTCHA, which requires potential clients to identify distorted text when creating an account online.
- Motivate why the company is implementing this software. (2)

- 6.4 An innovation corner was set up by CompuMart where customers can experience the benefits of virtual reality and augmented reality.
- 6.4.1 Describe TWO advantages of using augmented reality in business operations. (2)
- 6.4.2 A few customers at CompuMart have requested for a virtual reality (VR) setup for training applications.
- Discuss THREE key hardware requirements for running VR applications effectively and explain why these components are necessary. (3)
- 6.5 The store manager complains about the poor performance of the DBMS.
- 6.5.1 Give TWO reasons why data redundancy will have a negative impact on the efficiency of a DBMS. (2)
- 6.5.2 Suggest TWO ways to improve the efficiency of processing/filtering database queries. (2)
- 6.6 One of the consultants advised a customer to implement virtualisation on his current computer in order to develop code for iPhones.
- 6.6.1 Describe ONE way in which virtualisation can be used to reduce hardware costs for the customer. (2)
- 6.6.2 Explain TWO virtualisation features that enhance the process of recovering from disasters. (2)
- 6.6.3 Identify ONE challenge associated with implementing virtualisation. (1)
- TOTAL SECTION F: 28**
GRAND TOTAL: 150